MD. Rafi Imam

Real-Time Liveness Verification & Presentation Attack Detection System

Current Role: Software Development Officer (AI & Automation), Jamuna Bank PLC | **Degree:** B.Sc. ECE, RUET (First Class, 3.55 Last-60 GPA)
Contact: rafiimam96@gmail.com | **Portfolio:** rafiimam.github.io/Personal-website/ | **Target:** PhD Assistantship (Fall 2027)

1. EXECUTIVE SUMMARY & PRODUCTION SCOPE

This research profile details a full-stack, production-grade **Face Liveness Verification and Presentation Attack Detection (PAD) System**, independently engineered for high-security commercial banking infrastructure. Triggered during high-value transactions and automated risk flags, the platform executes a fail-closed three-layer verification cascade: (1) document-to-live face matching against enrolled ID media, (2) spatial and spatiotemporal multi-modal liveness verification, and (3) physics-informed presentation attack detection to defend against physical replay and digital injection spoofing. The core research focus addresses model survival under real-world mobile capture noise, non-rigid motion, and out-of-distribution (OOD) physical artifacts in unconstrained environments.

Core Systems & Algorithmic Highlights:
• **Decoupled Two-Tier Architecture:** Separates PyTorch CUDA inference (FastAPI REST) from a stateless, asynchronous orchestrator (asyncio.gather), reducing end-to-end evaluation latency from >5.0s to ~800ms.
• **Intermediate Feature Representation:** Hooks Stage-3 Block 5 in Swin-B Transformer to isolate mid-level texture anomalies (moiré, sub-pixel aliasing) from Stage-4 semantic identity features.
• **Physics-Informed Screen Replay Detection:** Combines column/row-FFT spectral periodicity analysis with Lucas-Kanade optical flow affine residuals to detect physical screen displays without relying solely on supervised classifiers.
• **Continual Adaptation via SALLoRA:** Integrates Sensitivity-Aware Layer-wise LoRA (PEFT) during background active-learning updates, using gradient sensitivity masking to adapt to emerging fraud patterns without model drift or full backbone retraining.

2. DISTRIBUTED SYSTEMS & ANTI-INJECTION INFRASTRUCTURE

Decoupled GPU Inference & Async Orchestration: Heavy neural network inference (MTCNN, Swin-B+UAD, ArcFace, 3D ResNet-18) is isolated on a dedicated FastAPI CUDA server. A stateless Gateway Orchestrator fires spatial, temporal, physics-based, and identity checks concurrently using Python's asyncio.gather(). Because modalities are conditionally independent given the video stream, parallel evaluation achieves real-time execution without sacrificing statistical rigor.

Vector Indexing & Cryptographic Anti-Injection: Enrolled NID document photos and signature card face templates are stored as 512-D ArcFace vectors in PostgreSQL leveraging the pgvector extension for sub-millisecond approximate nearest-neighbor (ANN) similarity search. To defeat digital stream injection (e.g., pre-recorded video or OBS virtual cameras), native client SDKs enforce cryptographic frame attestation, embedding HMAC-SHA256 signatures, session-bound nonces (TTL = 120s), and strict timestamp monotonicity checks across 8fps WebSocket JPEG streams.

3. ALGORITHMIC FORMULATIONS & COMPUTER VISION MODULES

A. Swin-B Transformer + Mid-Backbone Stage-3 Hook (Spatial PAD): To detect subtle physical presentation attacks, a custom Unified Attack Detection (UAD) head is attached via PyTorch register_forward_hook() at Stage 3, Block 5 of a Swin-B backbone. Stage-3 features retain granular spatial texture cues (print dot matrices, display refresh artifacts) that are typically lost at Stage 4 due to semantic identity abstraction. The UAD head incorporates a dual-path **HiLo Self-Attention** mechanism (62.5% local high-frequency window attention, 37.5% global low-frequency pooled attention) to simultaneously capture fine moiré fringes and global structural depth.

B. ArcFace Metric Learning & Document Face Extraction: Live face verification employs Additive Angular Margin Loss (ArcFace-R100) with scale $s = 64$ and angular margin $m = 0.5$ in normalized embedding space: $L_{Arc} = -\log \left[e^{s \cdot \cos(\theta_{y\_i} + m)} / \left(e^{s \cdot \cos(\theta_{y\_i} + m)} + \Sigma_{j \neq y\_i} e^{s \cdot \cos \theta\_j}\right) \right]$. A separate, lower-threshold MTCNN instance extracts degraded ID-photo faces from physical document scans, feeding 512-D embeddings into a 3-source fail-closed cascade (proprietary engine → cloud verification → internal ArcFace + pgvector).

C. 3D ResNet-18 Spatiotemporal Motion Modeling: To defeat static high-resolution printed photo attacks, a torchvision R3D-18 backbone (pretrained on Kinetics-400 and fine-tuned on fine-grained FAS sets) evaluates rolling 30-frame spatiotemporal chunks (batch × 3 × 30 × 112 × 112). Sharpness pre-filtering (Laplacian variance thresholding) discards motion-blurred frames prior to temporal convolution, yielding a robust stream-level fake score: $S_{temp} = 1 - mean(chunk\ scores)$.

D. Physics-Informed Screen & Replay Heuristics: Complementing learned models, a deterministic signal-processing engine exploits physical LCD sub-pixel properties:
• *RGB Sub-Pixel Striping:* Detects LCD column-periodic energy in horizontal pixel differences via multi-scale Column-FFT spectral correlation.
• *Non-Rigid Motion Residual:* Lucas-Kanade optical flow tracks 120 facial landmarks; a RANSAC-fitted affine residual isolates genuine non-rigid facial deformations from rigid planar screen movement.
• *Spectral Refresh Bands:* Inter-frame difference images are analyzed via Row-FFT with Hanning windowing to identify display refresh flicker.

4. SYNTHETIC DOMAIN GENERALIZATION & CONTINUAL PEFT (SALLORA)

SPSC (Simulated Physical Spoofing Clues): A major bottleneck in Presentation Attack Detection is the domain gap between academic benchmarks (collected under controlled lighting) and in-the-wild mobile banking captures. To prevent shortcut learning, we introduced SPSC—a physics-informed data synthesis pipeline applying stochastic combinations of JPEG re-compression, Moiré resampling, and chromatic column shifts to 85% of training spoofs and mild sensor noise to 75% of real captures.

Continual Production Adaptation via SALLoRA Integration: To handle evolving fraud mechanisms without full backbone retraining, the background update loop incorporates **Sensitivity-Aware Layer-wise LoRA (SALLoRA)**. When active-learning sessions are labeled via the administrative security API, an asynchronous background worker triggers a 2-epoch micro-batch update on the CUDA server. By applying gradient sensitivity masking to low-rank adapters attached to the Swin-B and 3D ResNet backbones, the system fine-tunes only parameters responsive to genuine corruption signals. This decouples domain adaptation from semantic knowledge, enabling zero-downtime hot-reloading of calibrated production weights without catastrophic forgetting.

5. ARCHITECTURAL POSITIONING VS. ACADEMIC BASELINES

Dimension	Standard Academic Baselines (e.g., FAS-TD, CDCN)	This System (Production & Research Framework)
Architecture	Monolithic offline classification model	Decoupled GPU server + parallel async orchestrator
Backbone Hook	Final-layer feature extraction (Stage 4)	Stage-3 Block 5 hook (mid-level texture disentanglement)
Screen Detection	Learned spatial appearance features only	Physics-informed Column/Row-FFT + Lucas-Kanade optical flow
Identity Matching	Omitted (Liveness evaluation only)	3-Source Fail-Closed Cascade (ArcFace + pgvector ANN)
Model Updating	Static offline training on benchmark splits	Continual micro-batch updates via SALLoRA (PEFT)
Security & Stream	Unprotected frame files / benchmark folders	HMAC signatures, nonces, timestamp monotonicity, 8fps WS

6. ALIGNMENT WITH FALL 2027 PHD RESEARCH THRUSTS

Bridging Enterprise AI Engineering & Foundational Research: This project demonstrates my ability to bridge theoretical machine learning (attention mechanisms, metric learning, parameter-efficient adaptation) with heavy, fault-tolerant systems engineering (distributed inference, vector databases, MLOps pipelines). In preliminary pilot sanity checks on production bank captures, the system established zero false accepts/rejects, serving as an operational baseline for ongoing benchmark collection. I aim to leverage this dual-threat foundation to pursue doctoral research in **Robust Multimodal Foundation Models, On-Device/Efficient Adaptation, and Trustworthy Vision Systems**.